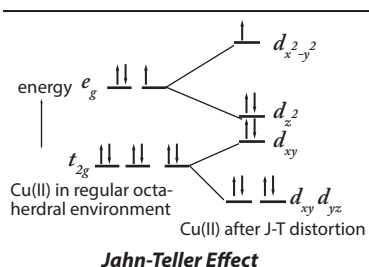


JAHN-TELLER EFFECT: A distortion of a highly symmetrical molecule, which reduces its symmetry and lowers its energy. The effect is observed in inorganic complexes and occurs for all nonlinear molecules in degenerate electronic states. When a structure of a nonlinear molecule or ion has degenerate orbitals, the actual structure of the molecule or ion is distorted so as to split the energy levels or increase the degeneracy. Degenerate orbitals are two molecular orbitals with the same energy levels.



JAMES SIX (1731 - 1793): British born scientist, who invented a deep sea self-registering thermometer in 1782 to measure and record maximum and minimum temperatures, named the Six's maximum and minimum thermometer, also known as maximum and minimum thermometer.

JAMES-LANGE THEORY: In psychology, it is a theory by William James (1842-1910), an American philosopher and psychologist Carl Georg Lange (1834-1900), a Danish physician that emotions are bodily sensations resulting from physiological responses to external stimuli when the brain reacts to the information received through the body's nervous system.

JAMES' THEOREM: The result that a subset of a Banach space is compact in the weak topology if and only if it is weakly closed and every continuous linear functional achieves its supremum on the set. In particular, a Banach space is reflexive if and only if every continuous linear functional supports the unit ball at some point.

JAMMING: In mathematics, the behaviour of a minimisation algorithm whose convergence is not global and which may therefore converge to a point that is not even a stationary point of the function in question.

JANET (JOINT ACADEMIC NETWORK): In computing, a wide area network linking academic and research institutes in the UK, operated by UKERNA (United Kingdom Education & Research Networking Association), a non-profit group, under contract from the Joint Information Systems Committee (JISC) of the UK Higher Education Funding Councils. It is controlled by a committee called the Joint Network Team and Network Executive (NE).

JANUS KINASE (JAK): A class of receptor-associated tyrosine kinases that transduce cytokine-mediated signals via the JAK-STAT pathway.

JASMONATE (JAs): A group of plant hormones, consisting of jasmonic acid and its esters such as methyl jasmonate (MeJa). They are synthesised from the unsaturated fatty acid linolenic acid and accumulate in injured parts of plants. Jasmonates play a role in the plant's defensive response against attack by insects, fungi and other pathogens. They are also involved in the regulation of plant growth and development.

JASPER: A red, yellow, brown or green coloured hard compact variety of chalcedony (SiO₂). It is formed from quartz. Good grades are used for semiprecious gemstones.

JATROPHA: A genus of flowering plants in the spurge family, Euphorbiaceae with about 170 species, native to the Americas but also found in Asia and Africa. They are mostly shrubs and small trees. Plants produce separate male and female flowers. The plants contain compounds that are highly toxic. Jatropha species have traditionally been used in basketmaking, tanning and dye production. The seeds of one species, *Jatropha curcas* (English common names include physic nut, Barbados nut, and poison nut) is used for biodiesel production.

JAUNDICE (ICTERUS): A yellow colouring of the skin and the whites of the eyes and mucous membranes. It is caused by deposition of

bile salts in these tissues. It occurs as a symptom of various diseases, such as hepatitis, that affect the processing of bile.

JAVA: A high-level programming language developed by Sun Microsystems with many similarities to C++. It is designed for creating web applications, that can operate on different platforms.

JAVA 3D: In computing, it is an application programming interface built into the latest version of the Java programming language that allows developers to write three-dimensional graphics applications and applets.

JAVA APPLET (APPLET): A program which can be distributed as an attachment in a World-Wide Web document and executed in a Java-enabled web browser. Web browsers can also be extended with plug-ins so that the applets can be run.

JAVA BEANS: An object-oriented programming interface devised by Sun Microsystems with which object-oriented programs can be created in the Java language. A Java Bean is similar to an ActiveX control in Microsoft's software architecture.

JAVA RUNTIME ENVIRONMENT (JRE; JAVA RTE): A software developed by Sun Microsystems that includes the Java Virtual Machine (JVM), code libraries and components, which are necessary to run programs written in Java. The JRE is available for multiple computer platforms, including Mac, Windows and Unix.

JAVA VIRTUAL MACHINE (JVM): A software implementation of a central processing unit (CPU) that runs compiled Java code (applets and applications).

JAVASCRIPT: A scripting language developed by Netscape to enable Web authors to design interactive sites and is supported in all Web browsers and other Web tools. It shares many of the features and structures of the full Java language but was developed independently.

JAW: Any of the two bony structures that form the framework of the mouth in all vertebrates except lampreys and hagfishes. In humans it consists of **mandible**, which is the movable lower jaw and the **maxilla**, which is the fixed upper jaw.

JAYNES-CUMMINGS MODEL (JCM): A theoretical model, put forward by the American physicists Edwin Jaynes (1922-98) and Fred Cummings in 1963, used in quantum optics and atomic physics to describe the system of a two-level atom interacting with a quantised mode of an optical cavity, with or without the presence of light. It has proved very useful in establishing which aspects of quantum optics are purely quantum mechanical and which can be dealt with by using quantum mechanics for the two-level atom and classical electrodynamics for the electromagnetic field.

JEANS INSTABILITY: Instability in interstellar gas clouds and subsequent star formation due to fluctuations in the density of the gas, causing the matter in the cloud to clump together and lead to gravitational collapse. The cloud is unstable if it is either very massive at a given temperature or very cool at a given mass for gravity to overcome the gas pressure.

JEANS LENGTH: The critical size of a gas cloud, of a given temperature and density, above which the cloud must collapse under its own gravity. Below this size, the cloud's internal pressure is enough to resist collapse.

JEANS MASS: The critical mass a volume of space must contain before it will collapse under the force of its own gravity.

JEJUNUM: A portion of the mammalian duodenum that lies between the duodenum and the ileum in higher vertebrates (reptiles, mammals and birds). Its main function is absorption of nutrients from digested food.

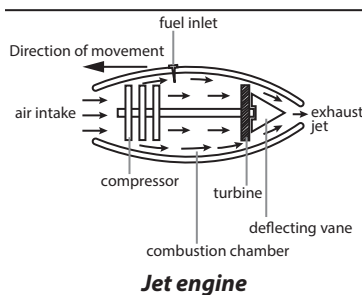
JELLYFISH: A marine invertebrate living in all oceans, belonging to the cnidarian phylum (coelenterates) with a radial symmetry body, with no head, a gut but no anus and a nervous system without a brain. Its umbrella-shaped body is composed of a semi-transparent gelatinous substance. It has tentacles covered with stinging cells (cnidocytes) that can sting or kill other animals. Most jellyfish use them to secure prey or as a defence mechanism. Some jellyfish such as Rhizostomae, do not have tentacles at all.

JENNET: 1. A female donkey or mule. 2. A small Spanish riding horse.

JENSEN INEQUALITY: 1. Named after the Danish mathematician Johan Jensen (1859 – 1925), it is an inequality involving convexity of a function. It is of the form $f(\sum \lambda_i a_i) \leq f(a)$, whenever $\sum \lambda_i = 1$ and $\lambda_i \geq 0$ that is satisfied by all convex combinations of points in the domain convex function and is equivalent to the convexity of the function f . 2. Various corresponding integral inequalities, often phrased in terms of probability.

JET: 1. A strong narrow stream of fluid, gas or steam forced out of a small nozzle. 2. An aircraft powered by a gas turbine (jet engine). 3. A hard black variety of lignite, a type of coal which is cut and polished for use in jewellery and ornaments. 4. In astronomy, a collimated stream of matter which flows from a central source. The jets can be broadly divided into two main types: **galactic jets**, which are believed to have a single source, a supermassive black hole at the centre of a galaxy; and **stellar jets**, which have several origins, for example, planetary nebulae, neutron stars and stellar black holes.

JET ENGINE: An internal-combustion engine such as turbojets, turboprops, turbopumps, turboshafts and ramjets that propel aircraft by means of the discharge of a jet of fluid towards the back. Air is drawn in, mixed with fuel and burnt. The resultant exhaust gases are forced out at very high speed to provide motion, based on the principles of conservation of momentum, so that the moving object is propelled in the opposite direction to the jet.



JET PROPULSION: Moving an object such as a rocket or airplane in one direction by passing a jet of fluid such as air or water in the opposite direction to the direction of motion. The motion is produced by the principles of conservation of momentum, so that the moving object is propelled in the opposite direction to the jet. Certain animals such as cephalopods, arthropods and fish have jet propulsion mechanisms for locomotion.

JET STREAM: A narrow fast-moving wind current that occurs in the Earth's atmosphere above the lower troposphere. The jet stream varies from about 161 to 644 km wide and 1.6 to 4.8 km thick. Its strongest winds are generally encountered at about 9,144 m in altitude in the troposphere. Jet-stream winds usually have a speed of 241 to 482 km/h. It may significantly influence aircraft operations because of the great speed of the wind at the jet core and the rapid spatial variation of wind speed in its vicinity.

JEWELLER'S ROUGE (IRON(III) OXIDE): A mild abrasive inorganic compound, which is a red powdered haematite, with the chemical formula Fe_2O_3 . It is one of the three main oxides of iron. The other two are iron(II) oxide (FeO) and iron(II,III) oxide (Fe_3O_4), which also occurs naturally as the mineral magnetite. Jeweller's rouge is used in metal cleaners and polishes.

JIGGER: A small metal cup used in measuring alcohol.

JMOL (JUDGMENT AS A MATTER OF LAW): A free, open source molecular program similar to RasMol used as an applet in a web page for students, educators and researchers in chemistry and biochemistry. It is cross-platform, running on Windows, Mac OS X and Linux/Unix systems.

JOHNSON SOLID: Any convex polyhedron with regular faces that is not a Platonic solid, an Archimedean solid or a prism (or an antiprism). There are 92 Johnson solids, which are named after Norman W. Johnson (1930 –), an American mathematician, who was the first to catalogue them in 1966. They include equilateral deltahedra and dipramids (two pyramids placed symmetrically base to base) and any irregular convex solid that can be made by sticking together triangles, squares, pentagons and so on, in a way that happens to close.

JOHNSON'S THEOREM: It states that if three congruent circles all intersect in a single point, then the other three points of intersection will lie on another circle of the same radius.

JOIN: In mathematics, in a partially ordered set, the join of a subset is the supremum of the subset. Given a pair of elements x and y their join, denoted $x \vee y$, is the element m such that $m \geq x$, $m \geq y$ and there is no $n < m$ that has the same relations to x and y .

JOIN PROBLEMS: Change problems where the change amount is "added" to the initial amount. In these problems, the initial amount, change amount, or resulting amount are unknown.

JOINT: 1. A vertical crack usually several metres in length formed in a rock by compression.

2. A structure connecting two or more bones. It may be movable such as fluid-filled synovial joints and those between vertebrae ball and socket joints. The bones of movable joints are covered with smooth cartilage and are lubricated by synovial fluid. Immobile joints include the sutures of the skull. 3. The portion of a stem from which a leaf or branch arises. This is mostly seen on grass stems. An example is the node.

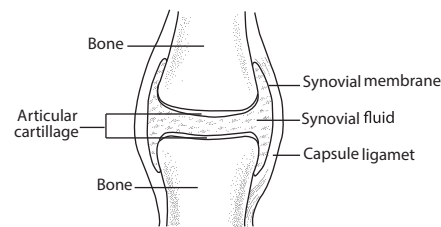


Fig. I: Vertical section through a synovial (movable) joint

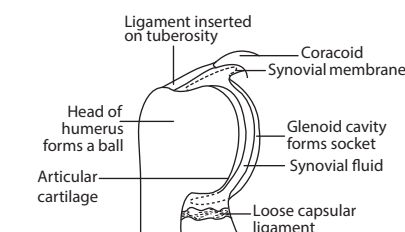


Fig. II: Section through the shoulder joint

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JOINT EUROPEAN TORUS (JET): A joint European research facility located at Culham Science Centre in Oxfordshire, UK that conducts experiments on nuclear fusion. It began operating in 1983 and is the focus of the European effort to produce a safe and environmentally sound fusion-power reactor. Its achievements include producing the world's first controlled release of deuterium-tritium fusion power in 1991; and in 1997 produced a world record 22 megajoules of nuclear fusion power by fusing isotopes of deuterium and tritium to produce 16 megawatts.

JOINT DISTRIBUTION FUNCTION: The joint distribution function of a pair of random variables is defined as the probability that the first random variable takes a value inferior or equal to the first real number and that simultaneously, the second variable takes a value inferior or equal to the second real number. Suppose X_1, X_2 are continuous random variables defined on the sample space and that:

$$F(X_1, X_2) = P(X_1 \leq x_1, X_2 \leq x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f(t_1, t_2) dt_1 dt_2 \quad \text{for all } x_1, x_2.$$

Then $f(x_1, x_2)$ is the joint density function of (X_1, X_2) provided $f(x_1, x_2) \geq 0$ and that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2 = 1$.

JOINT PHOTOGRAPHIC EXPERTS GROUP (JPEG): A graphics file format for editing still images, as well as colour facsimile, desktop publishing, graphic arts and medical imaging. JPEG is a joint standard of the International Telecommunications Union (ITU-T T.81) and the International Organization for Standardization (ISO 10918-1). Together with GIF (Graphics Interchange Format), they are the two

most commonly used file formats for pictures on the World Wide Web because pictures saved in these formats take up relatively small amount of space and help greatly reduce their file size.

JOINT PROBABILITY FUNCTION: A function that gives the probability that each of two or more random variables take a particular value.

JOINT VARIATION: A mathematical relationship which expresses how two or more variables vary with respect to another.

JOLIOT-CURIE, IRÈNE (1897-1956): French physicist, daughter of Marie and Pierre Curie, who discovered radium. She was educated by her mother and her scientist colleagues in physics; in mathematics by Paul Langevin; and in chemistry by Jean Baptiste Perrin. She later attended the Sorbonne. In 1921, she began to work at the Radium Institute, becoming director in 1946. In 1926, she married French nuclear physicist, Frédéric Joliot (1900-58). In 1935, they shared the Nobel Prize for chemistry for their synthesis of new radioactive elements.

JOLY'S STEAM CALORIMETER: An apparatus used to measure the specific heat capacity of a gas at constant volume. It has two equal spherical containers that are suspended from the opposite ends of the arm of a balance. One of the spheres is evacuated and the other contains the sample gas. The whole apparatus is enclosed in a steam bath and the specific heat capacity of the sample gas is calculated from the difference between the masses of the water that condenses on each sphere.

JONQUIÈRE'S FUNCTION (POLYLOGARITHM): In mathematics, the special function related to logarithm that is defined for integral $n \geq 2$ and z in the unit disk and written as $\text{Li}_n(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^n}$, here Li_2 is called the **dilogarithm** and Li_3 is called the **trilogarithm**.

JORDAN ALGEBRA: Named after French mathematician, Marie Ennemond Camille Jordan (1838 – 1922), it is non-associative algebra, whose multiplication satisfies the commutative as $ab = ba$ and the Jordan identity as $(a^2b)a = a^2(ba)$.

JORDAN BLOCK (JORDAN FACTOR): A Jordan block with value λ is a square matrix, whose entries are all λ on the diagonal, all 1 on the entries immediately above the diagonal and 0 elsewhere. It is of the form $J(\lambda) = S + \lambda I$, where λ is a scalar and S is super diagonal with entries equal to one.

JORDAN CONTENT (JORDAN-MEASURE; PEANO-JORDAN MEASURE): In mathematics, a form of measure especially useful in surface integration. It is an extension of the notion of size (length, area, volume) to shapes more complicated than, for example, a triangle, disk or parallelepiped. It is to the Riemann integral what measure is to the Lebesgue integral and provides the most practical definition of area.

JORDAN CURVE (JORDAN CONTOUR; SIMPLE CLOSED CURVE): A curve, such as a circle, whose shape is closed by a curved line or line-segments, does not intersect itself and has no end points. Every Jordan curve separates the plane into exactly two components.

JORDAN CURVE THEOREM: A theorem that every Jordan curve separates the plane into exactly two components: a bounded interior region (bounded by the curve) and an unbounded exterior region that contains all of the nearby and far away exterior points, so that any continuous path connecting a point of one region to a point of the other intersects with that loop somewhere.

JORDAN DECOMPOSITION: The Jordan decomposition of a signed measure is the expression of such a measure as the difference of two non-negative measures. Often these two measures are required to be mutually singular and then uniquely define the positive and negative parts of the signed measure.

JORDAN ELIMINATION (GAUSS-JORDAN ELIMINATION): A method of solving linear equations by row operations until an augmented matrix is transformed into a row echelon or reduced row-echelon form.

JORDAN MATRIX: A matrix whose diagonal elements are all equal and nonzero. Its elements above the principal diagonal are equal to 1, but all other elements are 0.

JORDAN NORMAL FORM: Of a matrix, a matrix similar to a given one and expressible in block diagonal matrix form as $\text{diag}[J(\lambda_1), \dots, J(\lambda_k), \dots, J(\lambda_r)]$, where each $J(\lambda_k)$ is a Jordan block. This representation is unique up to permutation of the blocks. If the scalar field is not algebraically closed a more complicated form results. Over the reals it is necessary to introduce factors corresponding to (real) irreducible quadratic factors of the characteristic polynomial.

JORDAN PRODUCT: The symmetric product of two quantities or operators A and B defined by $A \cdot B = 1/2(AB + BA)$.

JORDAN-HOLDER THEOREM: The theorem that for any two composition series of a finite group, an isomorphism exists between the corresponding quotient groups of each series, taken in some specified order.

JORDAN, MARIE ENNEMOND CAMILLE (1838-1922): French mathematician and engineer, who did research in topology, analysis and group theory and also made contributions to many mathematical fields. Some mathematical terms have been named after him including Jordan algebra, Jordan block (Jordan factor), Jordan curve (Jordan contour), Jordan curve theorem, Jordan decomposition, Jordan elimination (Gauss-Jordan elimination), Jordan matrix, Jordan measure (Jordan content), Jordan normal form, etc.

JOSEPHSON JUNCTION: Named after English physicist Brian Josephson, it is an electrical device consisting of two superconducting materials, which are separated by a thin layer of insulator, so that electrons can flow through the insulating barrier in the absence of an applied voltage, an effect known as **Josephson tunnelling**. It operates at temperatures close to absolute zero. Current flow between the superconductors in the absence of an applied voltage is called a **Josephson current**. Two or more of junctions joined by superconducting paths is called **Josephson interferometer**.

JOSEPHSON'S EFFECTS: Electrical effect associated with the tunnelling of electron pairs across an insulating barrier separating two superconductors.

JOULE: 1. The SI unit of work and energy, named after James Prescott Joule (1818-89), British physicist. It is defined as the amount of work done or the energy transferred by a force of one newton acting over a distance of one metre. Its symbol is j . 2. In the definition of volt, it is work required to move an electric charge of one coulomb through an electrical potential difference of one volt; or one *coulomb volt* ($C \cdot V$). 3. In the definition of watt, it is the work required to continuously produce one watt of power for one second; or one *watt second* ($W \cdot s$). 4. The S.I. unit of food energy or energy derived from food through the process of cellular respiration, which can be converted into calories as follows: $1 J = 0.239 \text{ cal}$ and $1 \text{ cal} = 4.184 J$. The calorie, which is also a metric system unit has been superseded by the S.I. unit joule, although it is still widely used.

JOULE HEATING (OHMIC HEATING; RESISTIVE HEATING): The process of heat production in a conductor as a result of the passage of an electric current through it. The heat produced is proportional to the square of the current multiplied by the electrical resistance of the wire: $J = I^2 R$. Its applications include incandescent light bulb, which glows when the filament is heated by Joule heating, so hot that it glows white with thermal radiation. All electric heaters such as electric stoves and others usually work by Joule heating.

JOULE METER: A device used to measure the electrical energy transfer in a circuit. An example is the meter used in houses to display electrical energy taken from the mains. Electric meters are typically calibrated in billing units, the most common one being the kilowatt hour.

JOULE-THOMSON EFFECT (JOULE-KELVIN EFFECT): The fall in temperature that occurs when a gas expands through a porous plug

into a region of lower pressure so that no heat is exchanged with the environment. For most real gases the temperature falls under these circumstances as the gas has to do internal work in overcoming the intermolecular forces to enable the expansion to take place. It can be felt when, for example, compressed air escapes through the valve of an inflated bicycle tyre. It is the basic principle of some refrigerators. At room temperature, all gases except hydrogen, helium and neon cool upon expansion by the Joule–Thomson process.

JOULE, JAMES PRESCOTT (1818-89): British physicist, who in 1840 discovered the relationship between electric current passing through a wire, its resistance and the amount of heat produced, known as Joule's law. In 1843, he published the amount of work required to produce a unit of heat, called the mechanical equivalent of heat and established that heat is a form of energy. In 1849, he gave an account of the kinetic theory of gases. He established that the various forms of energy are basically the same and can be changed from one into another, a discovery that formed the basis of the law of conservation of energy, the first law of thermodynamics. With William Thompson (Lord Kelvin), he discovered the Joule-Thompson effect. The value of the mechanical equivalent of heat is usually represented by the letter J and a standard unit of work is called the joule, named in his honour.

JOULE'S LAW (JOULE EFFECT; JOULE'S FIRST LAW): A physical law expressing the relationship between the heat generated by the current flowing through a conductor, named after James Prescott Joule (1818-89), British physicist. It states that the internal energy of a given mass of gas is independent of its volume and pressure. The heat (Q) produced when an electric current (I) flows through a resistance (R) for a time (t), is given by $Q = I^2 R t$. It is a function of temperature alone. This law applies only to ideal gases as in a real gas intermolecular forces would cause changes in the internal energy should a change of volume occur.

JOURNAL: 1. A cylindrical section of a shaft designed to rotate inside a bearing. 2. A periodical or magazine published, especially by a professional body or specialist containing contributions or information relevant to the area of activity of members of such professional body. 3. In computer science, a chronological record of changes made in a set of data; the record may be used to reconstruct a previous version of the set.

JOURNALING: The process of recording changes made in a physical file member in a journal. Journaling allows the programmer to reconstruct a physical member by applying the alterations made in the journal to a saved version of the physical file member.

JOVIAN: Relating to the planet Jupiter. A Jovian planet, also called a Gas giant, is used to describe Jupiter, Saturn, Uranus and Neptune.

JOYSTICK: An input device consisting of a lever used to control the display on a screen and used especially for games, flight simulators, etc. Joysticks consist of a base and a stick that can be moved in any direction. The stick can be moved slowly or quickly and in different amounts. Because of the flexible movements a joystick allows, it can provide much greater control than the keys on a keyboard.

JPEG (JOINT PHOTOGRAPHIC EXPERTS GROUP): A graphics file format for editing still images, as well as colour facsimile, desktop publishing, graphic arts and medical imaging. JPEG is a joint standard of the International Telecommunications Union (ITU-T T.81) and the International Organization for Standardization (ISO 10918-1). Together with GIF (Graphics Interchange Format), they are the two most commonly used file formats for pictures on the World Wide Web because pictures saved in these formats take up relatively small amount of space and help greatly reduce their file size.

JUGATE: Having parts that are arranged in pairs, as seen in the leaflets of a pinnate leaf.

JUGULAR VEIN: A pair of veins, the internal jugular and the external jugular veins found in the neck of mammals. The largest vein in the throat, it returns deoxygenated blood from the head to the superior vena cava and from there to the heart.

JUICE: The liquid from fruits such as orange and guava juices and also from vegetables or animal tissue.

JUKEBOX: A coin-operated hardware mechanism that plays any one of a group of music compact discs (CD) that is selected.

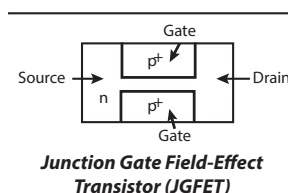
JUMP: 1. In mathematics, also known as saltus, the absolute value of the differences between the left- and right-hand limits of a given function (usually a function of bounded variation) at an interior point of its domain $|f(x+) - f(x-)|$. At an endpoint, the appropriate limit is compared to the function value, that is the jump at the endpoints of $]a, b[$ is respectively $|f(a+) - f(a)|$ and $|f(b) - f(b-)|$. 2. In computer science, a programming instruction that causes a computer to move to a different part of a programme rather than execute the next instruction in the program.

JUMP DISCONTINUITY: A point in a mathematical function where the left and right-hand limits exist but are not equal to each other.

JUMPER: In computer science, it refers to a small metal connector that acts as an on/off switch and is used to alter hardware configurations. Multiple jumpers, referred to collectively as a jumper block, are often used to tell the computer how a certain device, such as a hard drive or a modem, is configured. They can be found on motherboards, sound cards, graphics cards, I/O cards, CD-ROM interface boards, modems and hard drive controller boards and others.

JUNCTION DETECTOR (SOLID-STATE DETECTOR): A sensitive detector of ionising radiation in which the output is a current pulse proportional to the energy falling in or near the depletion region on a reverse-biased semiconductor junction.

JUNCTION FIELD-EFFECT TRANSISTOR, JFET (JUNCTION GATE FIELD-EFFECT TRANSISTOR, JUGFET): A type of field effect transistor (FET) in which the conducting channel lies between pn junctions in the silicon material. A pn junction acts as a diode, that becomes conductive if the gate voltage is reversed. It is the simplest type of field effect transistor (FET).



FET is one of the two major categories of transistors; the other is called bipolar junction transistors (BJTs), used as a switch or as a voltage-controlled resistance. The other type of field effect transistor (FET) is called the **insulated-gate field-effect transistor (IGFET)**, also known as the **metal oxide field effect transistor (MOSFET)**. Insulated gate field-effect transistors are unipolar devices just like JFETs and differs from the JFET by the addition of a silicon dioxide layer over the JFET and then a layer of silicon nitride. The result is a device which has even higher input impedance. The JFET consists of a semi-conductor material flanked by two highly doped layers of opposite type. If a positive voltage is applied to the drain, electrons in the source are attracted into the drain and flow through the channel between the source and drain. As the drain voltage increases, depletion layers associated with the p-n junction increases in size and reduce the cross-sectional area of the channel thus increasing the resistance of the device. When the depletion layers first meet, the drain voltage is called the pinch-off voltage. At drain voltages above pinch off, the depletion layers extend further into the drain region, but remain essentially constant in the channel. The current therefore remains substantially constant, until the voltage reaches such a value that breakdown occurs. Electrons arriving at the pinch-off point, are swept across the depletion layer into the drain region. The resulting drain current characteristics are similar to that of a pentode valve. Applying a negative voltage to the gates will also increase the size depletion layers and hence the pinch

off condition will be reached at a lower drain voltage and a family of characteristics will be generated for different values of gate bias. The gate voltage is therefore used to modulate the channel conductivity.

JUNCTION TRANSISTOR (BIPOLAR JUNCTION TRANSISTOR, BJT; BIPOLAR TRANSISTOR): A transistor formed by two P-N junctions, forming three regions, which functions to amplify an electric current. The two outer regions are called the **emitter** and **collector** and the middle region is called the **base**. These junctions are separated by a distance considerably less than a minority-carrier diffusion length, so that minority carriers injected at the emitter junction will not recombine before reaching the collector barrier and therefore, be effective in modulating the collector-barrier impedance. Junction transistors are classified as either NPN or PNP according to the arrangement of their N-type and P-type materials. An NPN transistor is formed by introducing a thin region of P-type material between two regions of N-type material. PNP transistor is formed by introducing a thin region of N-type material between two regions of P-type material. Silicon is the most widely used semiconductor material. Germanium is also used for some applications. Junction transistors are used both as discrete devices, found in the high-power and high-frequency applications and in integrated circuits.

JUNCTION UNIT: A non-repeating atom or non-repeating group of atoms between blocks in a block macromolecule.

JUNGLE: A popular name for rain forest. It is often covered with vegetation so dense that it is mainly impenetrable. It could also be described as a harsh environment characterised by fierce competitiveness or struggle for survival.

JUPITER: The fifth planet from the Sun and the largest in the solar system, with a mass of 1.8987×10^{27} kilograms, which is more than twice that of all the other planets combined. Earth has a mass of 5.9742×10^{24} kilograms. The surface area of Jupiter is 6.1419×10^{10} km². Its mean circumference is 4.39264×10^5 km, which is nearly 11 times larger than Earth's. It has a low density of 1.326 g/cm³, less than 25% of Earth's. It is composed of a ball of dense gas and has no solid surface. It is the fourth brightest object in the sky, after the Sun, the Moon and Venus. It is more than three times brighter than the brightest star, Sirius. It is visible to the naked eye, except for short periods when in near conjunction with the Sun. Jupiter orbits the Sun at an average distance of 778 million km, about five times the distance from Earth to the Sun. Jupiter's year or the time it takes to complete an orbit about the Sun, is 11.9 Earth years. Its day or the time it takes to rotate on its axis, is about 9.9 hours, less than half an Earth day.

JURASSIC: The second geological period of the Mesozoic era, following the Triassic extending over a period from about 200 million to 145 million years ago. It was named after the Jura Mountains on the borders of France and Switzerland by A. Brongniart in 1829.

JUSTIFICATION: 1. In mathematics, postulate, or theorem which enables a conclusion to be drawn. It includes showing or proving one's thinking or reasoning, for example, a solution, strategy, representation, or rationale pertaining to the situation. 2. In printing and word processing, the alignment of text to either the left or right margin or both. **Vertical justification** is adjusting the vertical space between lines so that columns and pages have an even bottom margin.

JUVENAL: The first covering of feathers on a bird, after it loses its down feathers.

JUVENILE: An individual organism that has not yet reached its adult form, sexual maturity or size, and in some cases, for example, in most animals cannot fend for itself.

JUVENILE HORMONE: A hormone secreted by insects from a pair of endocrine glands called the corpora allata close to the brain. It inhibits metamorphosis and maintains the presence of larval features in insects and is also important for the production of eggs in female insects.

JUVENILITY: The quality or condition of being juvenile, immature or being young. It is expressed in immature organisms, which are generally smaller in size than the adults and lack reproductive capability. In botany, seedling represents the juvenile stage of many plants.

JUXTAGLOMERULAR APPARATUS (JGA): A region of tissue found in each nephron in the kidney between the vascular pole of the renal corpuscle and the returning distal convoluted tubule of the same nephron. It is important in regulating blood pressure, body fluid and electrolytes. The juxtaglomerular apparatus has three cellular components: the macula densa, extraglomerular mesangial cells and juxtaglomerular cells or granular cells.

JUXTAMEDULLARY NEPHRON: A type of nephron found only in birds and mammals, but not in vertebrates with the renal corpuscle of the nephron situated close to or at the edge of the inner part of the kidney, called the medulla. It is one of the two major types of the kidney involved in urine production, the other one is the cortical nephron. The juxtamedullary nephrons have longer loops of Henle than the cortical nephron that extend far into the medulla and play an important part in the formation of urine. Each juxtamedullary nephron has a U-shaped network of blood vessels running alongside each loop of Henle, which are involved in the process of urine concentration, as they move sodium out into the medulla of the kidney. A nephron is the functional unit of the kidney, that regulates the concentration of water and soluble substances like sodium salts by filtering the blood, reabsorbing what is needed and excreting the rest as urine. There are about 800,000 to 1.5 million in each human kidney.

JUXTAPOSE: In mathematics, to multiply by placing factors side by side without a multiplication symbol ($\times$). At least one factor is a variable. For example, $5n$ means $5 \times n$, and ab means $a \times b$.



k: 1. The symbol for the metric prefix, kilo- with the value 1000 (10^3). For example, 1 kg (10^3) is equal to 1000 grams. 2. In mathematics, a vector on the z-axis, having a length of 1 unit. It is conventionally printed in bold type, **k**. 3. The symbol for Boltzmann constant, the ratio of the universal gas constant to Avogadro's number, with the value 1.3807×10^{-23} J/K.

K: 1. The symbol for the kelvin, the SI base unit of absolute or thermodynamic temperature. One kelvin represents the same temperature difference as one degree Celsius on the centigrade scale. 2. In mathematics, symbol for the special function, which represents the complete elliptic integral of the first kind with the formula

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta, \text{ which gives the period of a pendulum.}$$

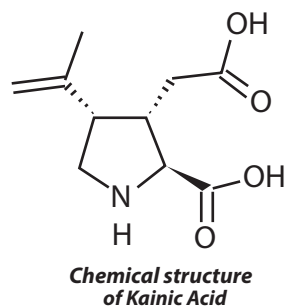
3. In computer science, the prefix kilo- with the symbol K- often represents 2^{10} , which is equal to 1024. This has been replaced by the prefix kibi- with the symbol Ki- in 1998 by a resolution of the International Electrotechnical Commission.

K-AR DATING (POTASSIUM-ARGON DATING): A method of isotopically dating the ages of rocks based on the radioactive decay of potassium-40 (^{40}K) to the stable isotope argon-40 (^{40}Ar). Ages are based on the known half-life of ^{40}K and the ratio of ^{40}K to ^{40}Ar . The method is used to measure a wide variety of ages of about 100,000 to 30 million years old.

K-SELECTED SPECIES (K-STRATEGIST SPECIES): Species whose populations increase until the populations begin to fluctuate at or near the carrying capacity. Some characteristics of *k*-selected species include the following: late maturation, fewer number, the young are larger and need more parental care, longer life spans and intense competition for resources. Examples of *k*-selected species include birds; large mammals such as elephants, horses and primates; and larger plants. They also tend to inhabit stable environments and are highly adapted to their environment.

KAGOSHIMA SPACE CENTRE: Now known as Uchinoura Space Centre, it is the launch site of Japan's Institute of Space and Astronautical Science (ISAS), situated at Kagoshima Prefecture on South Kyushu Island in a town called Uchinoura. Japan's first satellite, OHSUMI was launched from Kagoshima in 1970, becoming the 4th nation to do so after USSR, USA and France.

KAINIC ACID (DIGENIC ACID; ALPHA-KAINIC ACID; (2S,3S,4S)-3-(CARBOXYMETHYL)-4-PROP-1-EN-2-YLPYRROLIDINE-2-CARBOXYLIC ACID[1]; (3S,4S)-3-(CARBOXYMETHYL)-4-PROP-1-EN-2-YL-L-PROLINE; 2-CARBOXY-3-CARBOXYMETHYL-4-ISOPROPENYL-PYRROLIDINE): An analogue



of glutamic acid obtained from the red algae, *Digenaea simplex*. Its molecular formula is $\text{C}_{10}\text{H}_{15}\text{NO}_4$, density 1.22 g/cm^3 , boiling point 439.88°C at 760 mmHg, melting point $250\text{--}252.2^\circ\text{C}$. It is an amino-acid agonist for the kainite receptor, a neurotoxin and has anthelmintic properties.

KAINITE: A naturally occurring mineral salt of magnesium sulfate and potassium chloride, with the chemical formula $\text{MgSO}_4 \cdot \text{KCl} \cdot 3\text{H}_2\text{O}$. It is a source of potassium and, therefore, fertiliser.

KAINTE: Potash fertiliser made of a mixture of potassium and sodium salts, with sometimes magnesium salt added. It is used mainly on sugar beet and similar crops.

KAKEYA'S PROBLEM: The insoluble problem posed by Japanese mathematician, Soichi Kakeya (1886-1947) in 1917, of finding a solution to the smallest-area plane figure inside which a unit straight line segment can be rotated through 180° . In 1928, the Russian mathematician, Abram Besicovitch (1891-1970) showed that the problem had no answer or, in other words, there was no minimum area.

KAKUTANI FIXED POINT THEOREM: The theorem that every correspondence Γ that maps a compact convex subset C of a locally convex space into itself, with a closed graph and convex non-empty images (that is, $\Gamma(x)$ is a non-empty convex subset of C for all $x \in C$), has a fixed point, $x \in \Gamma(x)$. This extends Brouwer's theorem.

KALE (BORE COLE): A type of brassica belonging to the order Caparales. It is sometimes used as green vegetable for human consumption, but grown mainly as forage because of its nutritious green curled leaves.

KALUZA-KLEIN THEORY (KK THEORY): A type of theory that seeks to unify the two fundamental forces of gravitation and electromagnetism, named after Theodore Kaluza (1885-1954) and Oscar Klein (1894-1977). It postulates a generalisation of the theory of relativity to higher than four space-time dimensions. The theory further proposed that the higher dimensions are rolled up to become microscopically small with four microscopic space-time dimensions remaining. This process is known as **spontaneous compactification**.

KAM THEOREM (KOLMOGOROV-ARNOLD-MOSER THEOREM):

A theorem in dynamic systems about the persistence of quasi-periodic motions under small perturbations, important in celestial and statistical mechanics. It states that a non-negligible fraction of orbits in the phase space of a dynamical system remain indefinitely in a specific region of phase space even in the presence of small perturbations in the system.

KAME: A geological feature, which is usually in the form of a low, steep, rounded hill or ridge of layered sand and gravel drift, formed by the deposit of rocky materials carried by a stream of melting glacial ice.

KANGAROO: A name for a variety of mammals, classified as marsupials of the family Macropodidae, found in Australia, Tasmania and New Guinea. There are four species that include: (i) The red kangaroo (*Macropus rufus*), which is the largest marsupial in the world, with large males about 2 metres tall and weighing 90 kg; (ii) the western grey kangaroo (*Macropus fuliginosus*), with large males weighing about 54 kg; (iii) the antilopine kangaroo (*Macropus antilopinus*); and (iv) the eastern grey kangaroo (*Macropus giganteus*). Kangaroos are highly social animals and typically live in groups known as mobs or troops with 10 or more kangaroos. They have strong hind legs and conspicuously large feet with short forelegs and a long muscular tail for balancing when they hop. They cannot move one leg independently of the other and must therefore get around by hopping. The female has one young called **joey** at a time, which she